Tugas 3
Jaringan Kohonen

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TANGGAL	: Selasa, 25 Maret 2025

D4 TEKNIK KOMPUTER
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA
2025

TUGAS TEORI

Tugas :

Misalkan kita ingin mengcluster data-data berikut:

X_1	X_2
0,10	0,10
0,20	0,20
0,30	0,10
0,50	0,30
0,40	0,40
0,20	0,40

menjadi 2 cluster. Bobot awal yang akan kita gunakan adalah matriks berukuran 2x2 dengan tiap-tiap elemen bernilai 0,5. Learning rate (α=0,6) dengan tiap kenaikan epoh akan diset 0,5x(α). Maksimum epoh ditetapkan sebesar 10.

Program:

```
#include <stdio.h>
#include <math.h>

int main() {

float w[2][2] = {{0.5, 0.5},
{0.5, 0.5}};

float x[6][2] = {{0.10, 0.10},
{0.20, 0.20},
{0.30, 0.10},
{0.50, 0.30},
{0.40, 0.40},
{0.20, 0.40}};

float D[2];
int i, j, k, l;
float LR = 0.6;
int epoch_max = 10;

// Training
for (l = 0; l < epoch_max; l++) {
for (k = 0; k < 6; k++) {
for (j = 0; j < 2; j++) {
D[j] = 0.0;
for (i = 0; i < 2; i++) {
D[j] += (w[j][i] - x[k][i]) * (w[j][i] - x[k][i]);
}
}
printf("Data %.2f, %.2f: D[0] = %.4f, D[1] = %.4f\n", x[k][0],
x[k][1], D[0],
```

```
D[1]);

if (D[0] < D[1]) {
for (i = 0; i < 2; i++) {
w[0][i] += LR * (x[k][i] - w[0][i]);
}
} else {
for (i = 0; i < 2; i++) {

w[1][i] += LR * (x[k][i] - w[1][i]);
}
}
}
LR *= 0.5;
}
printf("\nHasil Pengelompokan:\n");
for (k = 0; k < 6; k++) {
for (j = 0; j < 2; j++) {
D[j] = 0.0;

for (i = 0; i < 2; i++) {
D[j] += (w[j][i] - x[k][i]) * (w[j][i] - x[k][i]);
}
}
printf("Data %.2f, %.2f: D[0] = %.4f, D[1] = %.4f\n", x[k][0],
x[k][1], D[0], D[1]);
if (D[0] < D[1]) {
printf("Data %.2f, %.2f berada di Cluster 1\n", x[k][0], x[k][1]);
} else {
printf("Data %.2f, %.2f berada di Cluster 2\n", x[k][0], x[k][1]);
}
}

return 0; }
```

Hasil training:

```
Data 0.30, 0.10: D[0] = 0.2000, D[1] = 0.0212
Data 0.50, 0.30: D[0] = 0.0400, D[1] = 0.0757
Data 0.40, 0.40: D[0] = 0.0104, D[1] = 0.0797
Data 0.20, 0.40: D[0] = 0.0577, D[1] = 0.0675
Data 0.10, 0.10: D[0] = 0.1265, D[1] = 0.0312
Data 0.20, 0.20: D[0] = 0.0479, D[1] = 0.0046
Data 0.30, 0.10: D[0] = 0.0881, D[1] = 0.0105
Data 0.50, 0.30: D[0] = 0.0510, D[1] = 0.0943
Data 0.40, 0.40: D[0] = 0.0029, D[1] = 0.0945
Data 0.20, 0.40: D[0] = 0.0294, D[1] = 0.0702
Data 0.10, 0.10: D[0] = 0.1287, D[1] = 0.0208
Data 0.20, 0.20: D[0] = 0.0481, D[1] = 0.0049
Data 0.30, 0.10: D[0] = 0.0811, D[1] = 0.0089
Data 0.50, 0.30: D[0] = 0.0398, D[1] = 0.1007
Data 0.40, 0.40: D[0] = 0.0037, D[1] = 0.0992
Data 0.20, 0.40: D[0] = 0.0244, D[1] = 0.0704
Data 0.10, 0.10: D[0] = 0.1315, D[1] = 0.0177
Data 0.20, 0.20: D[0] = 0.0494, D[1] = 0.0048
Data 0.30, 0.10: D[0] = 0.0791, D[1] = 0.0083
Data 0.50, 0.30: D[0] = 0.0348, D[1] = 0.1036
Data 0.40, 0.40: D[0] = 0.0039, D[1] = 0.1011
Data 0.20, 0.40: D[0] = 0.0225, D[1] = 0.0705
Data 0.10, 0.10: D[0] = 0.1330, D[1] = 0.0165
Data 0.20, 0.20: D[0] = 0.0502, D[1] = 0.0047
Data 0.30, 0.10: D[0] = 0.0783, D[1] = 0.0081
Data 0.50, 0.30: D[0] = 0.0326, D[1] = 0.1049
Data 0.40, 0.40: D[0] = 0.0039, D[1] = 0.1021
Data 0.20, 0.40: D[0] = 0.0217, D[1] = 0.0705
Data 0.10, 0.10: D[0] = 0.1338, D[1] = 0.0159
Data 0.20, 0.20: D[0] = 0.0506, D[1] = 0.0046
Data 0.30, 0.10: D[0] = 0.0779, D[1] = 0.0079
Data 0.50, 0.30: D[0] = 0.0315, D[1] = 0.1055
Data 0.40, 0.40: D[0] = 0.0039, D[1] = 0.1025
Data 0.20, 0.40: D[0] = 0.0213, D[1] = 0.0705
Data 0.10, 0.10: D[0] = 0.1342, D[1] = 0.0157
Data 0.20, 0.20: D[0] = 0.0508, D[1] = 0.0046
Data 0.30, 0.10: D[0] = 0.0778, D[1] = 0.0079
Data 0.50, 0.30: D[0] = 0.0310, D[1] = 0.1059
Data 0.40, 0.40: D[0] = 0.0039, D[1] = 0.1027
Data 0.20, 0.40: D[0] = 0.0211, D[1] = 0.0705
Data 0.10, 0.10: D[0] = 0.1344, D[1] = 0.0155
Data 0.20, 0.20: D[0] = 0.0509, D[1] = 0.0046
Data 0.30, 0.10: D[0] = 0.0777, D[1] = 0.0078
Data 0.50, 0.30: D[0] = 0.0307, D[1] = 0.1060
Data 0.40, 0.40: D[0] = 0.0039, D[1] = 0.1029
Data 0.20, 0.40: D[0] = 0.0210, D[1] = 0.0706
Data 0.10, 0.10: D[0] = 0.1345, D[1] = 0.0155
Data 0.20, 0.20: D[0] = 0.0510, D[1] = 0.0046
Data 0.30, 0.10: D[0] = 0.0776, D[1] = 0.0078
Data 0.50, 0.30: D[0] = 0.0306, D[1] = 0.1061
Data 0.40, 0.40: D[0] = 0.0039, D[1] = 0.1029
Data 0.20, 0.40: D[0] = 0.0209, D[1] = 0.0706
Data 0.10, 0.10: D[0] = 0.1345, D[1] = 0.0154
Data 0.20, 0.20: D[0] = 0.0510, D[1] = 0.0046
Data 0.30, 0.10: D[0] = 0.0776, D[1] = 0.0078
Data 0.50, 0.30: D[0] = 0.0306, D[1] = 0.1061
Data 0.40, 0.40: D[0] = 0.0039, D[1] = 0.1029
Data 0.20, 0.40: D[0] = 0.0209, D[1] = 0.0706
```

Hasil testing:

```
Hasil Pengelompokan:
Data 0.10, 0.10: D[0] = 0.1346, D[1] = 0.0154
Data 0.10, 0.10 berada di Cluster 2
Data 0.20, 0.20: D[0] = 0.0510, D[1] = 0.0046
Data 0.20, 0.20 berada di Cluster 2
Data 0.30, 0.10: D[0] = 0.0776, D[1] = 0.0078
Data 0.30, 0.10 berada di Cluster 2
Data 0.50, 0.30: D[0] = 0.0305, D[1] = 0.1061
Data 0.50, 0.30 berada di Cluster 1
Data 0.40, 0.40: D[0] = 0.0039, D[1] = 0.1029
Data 0.40, 0.40 berada di Cluster 1
Data 0.20, 0.40: D[0] = 0.0209, D[1] = 0.0706
Data 0.20, 0.40 berada di Cluster 1
```